PROJECT DEVELOPMENT PHASE

1. Introduction

The **Project Development Phase** focuses on building and implementing the system based on the requirements and design created in earlier phases.
In this phase, the functionalities for **data import, field mapping, and relationship mapping** are developed using the features provided by the ServiceNow platform.

The main goal is to create a **working system that automatically imports data and correctly maps relationships between database tables**.

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2. Development Environment

The development environment includes:

Component	Description
Platform	ServiceNow
Development Tool	ServiceNow Studio / Platform Interface
Data Format	CSV or Excel files
Browser	Chrome / Edge / Firefox

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3. System Modules

The project development consists of the following modules:

1. Data Import Module

This module allows administrators to upload external data files.

Functions:

- Upload CSV or Excel files
- Read file content
- Send data to Import Set table

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2. Import Set Configuration

The **Import Set** acts as a temporary storage for incoming data.

Tasks performed:

- Create Import Set table
- Configure Import Set Loader

- Store uploaded records in the staging table

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3. Transform Map Development

The **Transform Map** converts data from the Import Set table to the target tables.

Functions:

- Map source fields to target fields
- Apply transformation rules
- Ensure correct data format

Example mapping:

Source Field	Target Field
user_name	name
team_id	team
department_id	department

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4. Relationship Mapping using Dot Walking

Dot Walking allows the system to access fields from related tables.

Example:

user.team.department.name

This means:

- User → Team → Department → Department Name

This technique helps in **linking multiple related tables automatically**.

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5. Data Validation Module

This module checks the imported data before inserting it into the database.

Validation includes:

- Checking empty fields
- Detecting duplicate records
- Verifying correct data types

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6. Data Insertion Module

After validation, the processed data is inserted into the **target tables**, such as:

- User table
- Team table
- Department table

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4. Development Workflow

Data File Upload
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Import Set Loader
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▼
Import Set Table
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▼
Transform Map
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▼
Dot Walking (Relationship Mapping)
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▼
Data Validation
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Target Tables Updated

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5. Tools and Technologies Used

Technology	Purpose
ServiceNow Import Sets	Data import
Transform Maps	Field mapping
Dot Walking	Relationship mapping
ServiceNow Tables	Data storage

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6. Expected Output

After development, the system will:

- Automatically import external data.
- Map relationships between tables correctly.

- Reduce manual data entry.
- Improve data accuracy and efficiency.